

Home work

write a program to determine length of linked list
from prog2 import*

```
class sll(linked_list):
```

```
    def length(a):
```

```
        p = a.first
```

```
        cnt = 0
```

```
        while p != None:
```

```
            cnt + 1
```

```
            p = p.link
```

```
        return cnt
```

```
# End of the class
```

```
if __name__ == '__main__':
```

```
    a = sll()
```

```
    a.create()
```

```
    print('Number of nodes:', a.length())
```

write a program to determine data of ith node
from prog4 import*

```
class linkedlist(sll):
```

```
    def find(a, i):
```

```
        if i < 1 or i > a.length():
```

```

        return None
    p = a.first
    for j in range(i-1):
        p = p.link
    return p.data

# End of the class
a = linkedlist()
a.create()

while True:
    i = int(input("Enter value of i: "))
    x = a.find(i)
    if x == None:
        print(f'Node {i} does not exist')
    else:
        print(f'Data of node {i} is: {x}')
    ch = input('Do you wish to continue (y/n): ')
    if ch == 'N' or ch == 'n':
        break
# End of while loop
print('Good Bye')

```

Write a method to search for a value in the linked list

from prog4 import *

class sll(linked_list):

def search(a, x):

p = a.first

while p != None:

if p.data == x:

return p

else:

p = p.link

return None

End of the class

```
a = sll()
```

```
a.create()
```

```
while True:
```

```
    x = eval(input("Enter value to be searched: "))
```

```
    p = a.search(x)
```

```
    if p == None:
```

```
        print(f'{x} is not found')
```

```
    else:
```

```
        print(f'Found at: address: {p}') 
```

```
    ch = input('Do you wish to continue(y/n):')
```

```
    if ch == 'N' or ch == 'n':
```

```
        break
```

```
# End of while loop
```

```
print('Good Bye')
```

```
# write a method to insert a node in the linked list.
```

```
from prog4 import *
```

```
class linkedlist(sll):
```

```
    def insert(a, i, x):
```

```
        if i < 0 or i > a.length():
```

```
            print(f'Node {i} does not exist')
```

```
        elif i == 0:
```

```
            new = node(x)
```

```
            new.link = a.first
```

```
            a.first = new
```

```
        else:
```

```
            new = node(x)
```

```
            p = a.first
```

```
            for j in range(i-1):
```

```
                p = p.link
```

```
            new.link = p.link
```

```
            p.link = new
```

```
# End of the class.
```

```

a = LinkedList()
a.create()
while True:
    i = int(input("Enter value of i: "))
    x = eval(input("Enter value to be inserted: "))
    a.insert(i, x)
    a.display()
    ch = input("Would you like to insert another node  

    if ch == 'N' or ch == 'n': (Y or N)? : ")
    break.
# End

```

Write a method to delete i th node of linked list.

```

from prog4 import *
class singly_linked_list(sll):
    def delete(a, i):
        if i < 1 or i > a.length():
            return None
        elif i == 1:
            p = a.first
            x = p.data
            a.first = a.first.link
            del p
            return x
        else:
            p = a.first
            for j in range(i-2):
                p = p.link
            next = p.link
            p.link = next.link
            x = next.data
            del next
            return x
# End of the class

```

a = singly_linked_list()

a.create()

while True:

i = int(input('enter value of i: '))

x = a.delete(i)

if x == None:

print('Node {} does not exist')
i

else:

print('data of deleted node is', x)

a.display()

ch = input('would you like to delete another
node (y or N)? :')

if ch == 'n' or ch == 'N':

break.